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10 CFR 50.73

GNRO2023-00006

February 16, 2023

U.S. Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, DC 20555-0001

SUBJECT: Grand Gulf Nuclear Station, Unit 1 Licensee Event Report 2022-003-00,
Manual Reactor SCRAM due to trip of the A Reactor Feedwater Pump

Grand Gulf Nuclear Station, Unit 1
Docket No. 50-416
Renewed License No. NPF-29

Attached is Licensee Event Report 2022-003-00, Manual Reactor SCRAM due to trip of the A Reactor Feedwater Pump. This report is being submitted in accordance with 10 CFR 50.73(a)(2)(iv)(A) as an event or condition that resulted in manual or automatic actuation of the Reactor Protection System.

This letter contains no new Regulatory Commitments. Should you have any questions concerning the content of this letter, please contact Jeff Hardy, Regulatory Assurance Manager at 802-380-5124.

Sincerely,

A handwritten signature in black ink, appearing to read 'JAH' followed by a stylized flourish.

JAH/saw

Attachments: Licensee Event Report 2022-003-00

cc: NRC Senior Resident Inspector
Grand Gulf Nuclear Station
Port Gibson, MS 39150

U.S Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

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Page 3 of 3

Attachment
Licensee Event Report 2022-003-00

NRC FORM 366 (01-10-2023)		U.S. NUCLEAR REGULATORY COMMISSION			APPROVED BY OMB: NO. 3150-0104		EXPIRES: 12/31/2023				
 LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)											
1. Facility Name Grand Gulf Nuclear Station, Unit 1					☒ 050 ☐ 052		2. Docket Number 416		3. Page 1 OF 3		
4. Title Manual Reactor SCRAM due to a loss of the Condensate and Feed Water System											
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number	
12	19	2022	2022	003	00	02	16	2023	N/A	N/A	
									N/A	N/A	
9. Operating Mode 1						10. Power Level 78					
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)											
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		☐ 73.1200(a)	
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)		☐ 73.1200(b)	
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)		☐ 73.1200(c)	
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.36(c)(2)		☒ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		☐ 73.1200(d)	
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(A)		10 CFR Part 73		☐ 73.1200(e)	
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)		☐ 50.69(g)		☐ 50.73(a)(2)(v)(B)		☐ 73.77(a)(1)		☐ 73.1200(f)	
☐ 20.2203(a)(2)(iii)				☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(2)(i)		☐ 73.1200(g)	
☐ 20.2203(a)(2)(iv)				☐ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(ii)		☐ 73.1200(h)	
☐ 20.2203(a)(2)(v)				☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)					
☐ OTHER (Specify here, in abstract, or NRC 366A).											
12. Licensee Contact for this LER											
Licensee Contact Jeffery A. Hardy								Phone Number (include area code) 802-380-5124			
13. Complete One Line for each Component Failure Described in this Report											
Cause	System	Component	Manufacturer	Reportable To IRIS	Cause	System	Component	Manufacturer	Reportable To IRIS		
X	SD	NA	NA	Y	NA	NA	NA	NA	NA		
14. Supplemental Report Expected							15. Expected Submission Date		Month	Day	Year
☒ No		☐ Yes (If yes, complete 15. Expected Submission Date)									
16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)											
At 2058 on December 19, 2022, Grand Gulf Nuclear Station, Unit 1 experienced a trip of Condensate Booster Pump A. At 2101, Reactor Feedwater Pump A tripped, and the reactor was manually scrammed. All control rods were fully inserted into the core. At 2104, Reactor Feedwater Pump B tripped. At 2121 High Pressure Core Spray (HPCS) was manually injected to maintain reactor water level. At 2126 Reactor Feedwater Pump A was successfully restarted. Following the reactor SCRAM, the HPCS system was used to maintain reactor water level. The direct cause of the SCRAM was the Heater Drain Tank (HDT) Level Control Valves were not capable of responding quickly enough to mitigate the HDT level oscillations during the transient caused by Condensate Booster Pump Trip and subsequent Fast Detent.											
There were no consequences to the safety of the public, nuclear safety, industrial safety, or radiological safety. No radiological releases occurred due to this event. This report is made in accordance with 10 CFR 50.73(a)(2)(iv)(A) for any event or condition that resulted in manual or automatic actuation of any system listed in paragraph 10 CFR 50.73(a)(2)(iv)(B). The Reactor Protection System including reactor SCRAM or trip is included in 10 CFR 50.73(a)(2)(iv)(B).											

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collection Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; e-mail: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME

Grand Gulf Nuclear Station, Unit 1

☒ 050☐ 052**2. DOCKET NUMBER**

416

3. LER NUMBER

YEAR

2022

SEQUENTIAL
NUMBER

- 003

REV
NO.

- 00

NARRATIVE**PLANT CONDITIONS:**

Mode 1, 78 percent power

DESCRIPTION OF EVENTS

At 2058 on December 19, 2022, while the plant was operating at 100 percent reactor power, Condensate Booster Pump (CBP)[SD] A tripped. Control Room Operators entered the Feedwater System Malfunction Off-Normal Event Procedure (ONEP) and reduced core flow to 70 million pound-mass per hour. While performing steps of the ONEP operators identified Reactor Feedwater Pump "A" and "B" suction pressures were rapidly lowering. At 2101, Reactor Feedwater Pump "A" tripped on low suction pressure. A recirc flow control valve runback occurred as designed. Reactor water level continued to lower because of a reduction in feed flow. At approximately 17 inches of water and trending down, the At-The-Controls Operator placed the Reactor Mode Switch in Shutdown at 2101. At 2121 High Pressure Core Spray (HPCS) [BG] was manually injected to maintain reactor water level.

The manual SCRAM on low reactor water level resulted from constricting Feed Flow when the Heater Drain Tank Level Control Valves closed, which is a plant response that should have been found in the results of calculation for the "Condensate and Feedwater Transient Analysis" (CFTA). This oversight, along with other issues found in the CFTA calculation led to the Root Cause of "Transients caused by CBP Trip have not been adequately analyzed", and this condition will be corrected by performing an analysis to determine the best corrective modification and implementing that modification during the next refueling outage.

REPORTABILITY

This report is made in accordance with 10 CFR 50.73(a)(2)(iv)(A) for any event or condition that resulted in manual or automatic actuation of any system listed in paragraph 10 CFR 50.73 (a)(2)(iv)(B). The Reactor Protection System including reactor SCRAM or trip is included in 10 CFR 50.73 (a)(2)(iv)(B).

CAUSE

Direct Cause: Heater Drain Tank (HDT) Level Control Valves were not capable of responding quickly enough to mitigate the HDT level oscillations during the transient caused by Condensate Booster Pump Trip and subsequent Fast Detent.

Root Cause: Transients Caused by Condensate Booster Pump Trip have not been adequately analyzed.

CORRECTIVE ACTIONS

1. Replace CBP A Motor – The CBP A Motor has been replaced.
2. Corrective Action to Prevent Recurrence (CAPR) – Modification will be performed to implement long term solutions as determined to analyze the transient effects of the following on Feed Flow and Post Extended Power Uprate (EPU) and Maximum Extended Load Line Limit Analysis Plus (MELLA+) conditions: Condensate Pump trip, CBP trip, HDP trip. Develop potential solutions to solve latent design issues in Feed Flow transient response.



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1. FACILITY NAME Grand Gulf Nuclear Station, Unit 1	☒ 050	2. DOCKET NUMBER 416	3. LER NUMBER		
	☐ 052		YEAR 2022	SEQUENTIAL NUMBER - 003	REV NO. - 00

SAFETY SIGNIFICANCE

There were no consequences to the safety of the public, nuclear safety, industrial safety, or radiological safety. No radiological releases occurred due to this event.

Following the SCRAM, HPCS remained operable and was used to feed the reactor. Safety / Relief Valves were operable and available to reduce reactor pressure if required and safety-related Emergency Core Cooling Systems were operable.

The condensate and feedwater systems are described in the Section 10.4.7 of the Updated Final Safety Analysis Report (UFSAR). The function of the condensate and feedwater system is to provide a dependable supply of feedwater to the reactor, to provide feedwater heating, and to maintain high water quality in the feedwater. The system is designed to provide the required flow at the required pressure to the reactor, allowing sufficient margin to provide continued flow under anticipated transient conditions.

The condensate and feedwater system serves no safety function. Systems analysis has shown that failure of this system will not compromise any safety-related systems or prevent safe shutdown. The condensate and feedwater system is not required to effect or support the safe shutdown of the reactor or perform in the operation of reactor safety features.

PREVIOUSLY SIMILAR EVENTS

A review of internal operating experience for the previous three years identified the following similar event.

LER 2020-004-01 Automatic Reactor SCRAM Due to Reactor Feed Pump Trip